

AP CSP | Lesson 25: Fault Tolerance

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B describes useful alternatives. A overclaims; C reduces redundancy; D changes only presentation.

Q2. Answer: C

C ties a failure to a maintained function. A observes no fault; B is cosmetic; D invents immunity for software.

Q3. Answer: A

A uses the stated working edges. B uses failed A-C; C and D invent absent links.

Q4. Answer: D

D follows from no working first link. A and B invent connectivity; C ignores complete isolation.

Q5. Answer: A and D

A states a resource cost. D recognizes shared infrastructure. B assumes independence from a diagram; C overclaims end-to-end service. Select A and D only.

Q6. Answer: B

B respects missing link/waiting information. A invents timing units; C invents capacity; D invents delivery reliability.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Examples S-A-C-T and S-B-D-T, each 3 hops. The second avoids A-C and remains usable.

1 mark: Gives one valid S-to-T path.

1 mark: Gives a different valid path.

1 mark: Correctly counts hops for both.

1 mark: Identifies a stated route avoiding failed A-C.

Q8. 4 marks

Answer: S-A-C-T; 3 hops; it uses none of C-D; no, the failure can affect alternatives without changing this route.

1 mark: Gives a valid route avoiding C-D.

1 mark: Correctly counts that route's hops.

1 mark: Explains S-A-C-T excludes C-D.

1 mark: Distinguishes an unchanged selected path from absence of a failed component.

Q9. 4 marks

Answer: No; T has no working incident links. S-A and S-B is another disconnecting set. A route cannot start/end across a missing local connection.

1 mark: States no S-to-T path.

1 mark: Identifies isolation of T.

1 mark: Gives S-A and S-B as a valid alternative set.

1 mark: Explains that remote edges cannot repair the isolated endpoint connection.

Q10. 4 marks

Answer: S-A-C-D-T; 4 hops; 4 units; the equal-hop cost is an assumption and real links/queues differ.

1 mark: Gives S-A-C-D-T.

1 mark: Counts four hops.

1 mark: Calculates four stated time units.

1 mark: Identifies the timing assumption or omitted real delay factors.

Q11. 4 marks

Answer: Tunnel damage could affect both. Logical alternatives share physical infrastructure. Use a physically separate route; this can require extra cable, equipment or maintenance.

1 mark: Identifies a common tunnel-related failure.

1 mark: Explains shared infrastructure connects their failure risk.

1 mark: Proposes physical separation or another relevant independent alternative.

1 mark: Names a plausible resource/maintenance cost.

Q12. 4 marks

Answer: It supports reachability for that tested failure. S-A and S-B together disconnect. A failed receiver/application can still prevent service. The tested link failure was tolerated for S-to-T reachability.

1 mark: Limits evidence to the tested failure and communication function.

1 mark: Gives a valid disconnecting failure set.

1 mark: Identifies an endpoint/application or other omitted failure.

1 mark: States a bounded claim rather than immunity.